

Shower Device

Happy Campers

Element C: Presentation and Justification of Solution Design Requirements

Dormitory residents and dormitory guests of our boarding school need a device to place their shower products. This is because they would want their shower products to be easily accessible in the communal shower. They would also like to do this without having a personal shower caddy and without setting it on the floor because it is unsanitary. The main stakeholders are the dormitory residents. They need a solution because the showers in the dormitory were not built with a solution to this problem and dormitory residents are not permitted to leave their shower products in the shower area. The only place to put a device in the dormitory showers are on a relatively narrow divider wall of less than an inch wide.

The design for our device must meet several crucial criteria. List of priority from high to low:

- Waterproof (high priority)
- Elevated (connected to the shower divider at a spot where everyone can reach it) off the floor (high priority)
- Accommodate five bottles and a razor (high priority)
- Sturdy enough to support up to 10lb (medium priority)
- Lightweight (less than 2lb) (low priority)
- No deeper than 6 inches, and no wider than 12 inches (low priority)
- Portability and the ability to be easily removed with one hand (low priority)

In terms of constraints, it must be easily cleanable with standard supplies and stay within

a budget of \$10 per shower. There should be no issue with standing water, and the caddy must maintain balance and stability even when bumped into.

Our top priorities are a design that can hold all necessary shower items without taking up too much space in the shower. Water resistance, cost-effectiveness, and ample storage are of utmost importance. While lightweight, portability, and sturdiness are still significant factors, they are slightly lower in priority.

The \$10 per shower budget is critical, allowing us to manage costs effectively. Ensuring our design is waterproof will contribute to its longevity, reducing the need for repairs or replacements. Ultimately, if the caddy cannot accommodate all essential shower items, it would render its installation pointless.

Our stakeholders agree with the proposed design requirements because it fits their needs of the importance of affordability and user-friendliness. Specifically the stakeholders who use the dormitory showers everyday were happy to see another option to the current situation. Our stakeholders are the dorm residents (the ones using our shower device), dorm parents (deal with the overall living conditions of the dorm), maintenance staff (the people cleaning the dorm showers), and parents of the dorm residents (no longer have to buy a shower caddy every year).